Tangkabati 1 Watershed

B-76

Figure 1: Para (village) boundary of Tangkabati 1 watershed

Source:
Center for Environmental and Geographic Information Services (CEGIS)

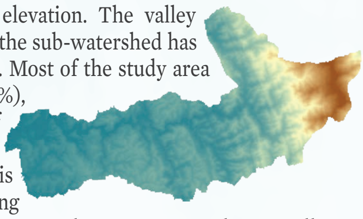
Disclaimer: Boundaries are not necessarily authoritative

The Tangkabati 1 sub-watershed (B-76), located in the Tangkabati Union of the Bandarban Sadar Upazila of the Bandarban District, encompasses 12 paras (figure 1) and covers approximately 2,549 hectares of land. It is home to over 2,500 people, with communities consisting of the Mro, Non-IP, and Tripura ethnic groups. The overall literacy rate is approximately 63%, yet opportunities for higher education are very limited. Farming is the main occupation. The average monthly income is 12,947 taka ($108), which is relatively low compared to the national average. About 87% of the houses are katcha and almost 85% of households use solar electricity systems. There is a noticeable scarcity of drinking water and a prevalent incidence of waterborne diseases.

Topography

Only 10% area lies below 50m elevation. The valley situated in the western regions of the sub-watershed has the lowest elevation in the region. Most of the study area lies between 50-150m (60%), highlighting the dominance of moderately low-lying areas in the watershed. The next largest range is 150–250 meters, comprising approximately 14% of the area. Elevation above 250 meters have small portion of land (nearly 15%). The higher elevation areas (>550m) are in the eastern part of the watershed.

Figure 2: Topography



Source:
Center for Environmental and Geographic Information Services (CEGIS)

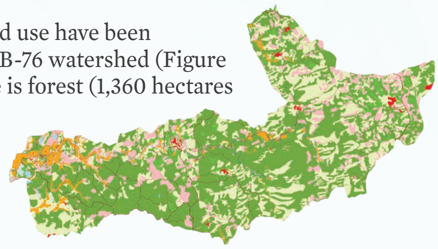
Disclaimer:
Boundaries are not necessarily authoritative

meter Mean Sea Level (mMSL)

Landuse

A total 13 types of land use have been identified within the B-76 watershed (Figure 3). The main land use is forest (1,360 hectares (ha), followed by herb/shrub (646 ha), shifting cultivation (272 ha), and orchard (104 ha).

Figure 3: Land use



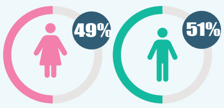
- Forest Tree Dominated Area (53%)
- Crop Land (4%)
- Herb/Shrub Dominated Area (25%)
- Rural Settlement (1%)
- Shifting Cultivation (11%)
- Orchard (4%)
- Others (2%)

Disclaimer:
Boundaries are not necessarily authoritative

Source:
Center for Environmental and Geographic Information Services (CEGIS)

Demography

Population



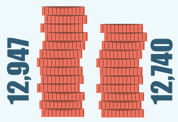
Total 2,624

Literacy

Male 67% Female 58%

Average literacy rate 62%

Economy

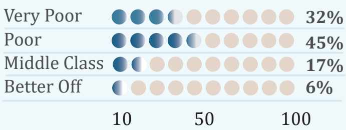


Monthly Income (BDT) Monthly Expense (BDT)

WASH

- 28 tubewell, 8 gravity feed
- 411 sanitary latrines
- 14 dug wells, 1 ring wells
- 5 chharas (creeks), 6 springs

Poverty



Production

- Rice 5-6 metric tons per hectare (MT/acre)
- Fisheries 4-5 MT/acre
- 5,119 Livestock

Climatic Hazards

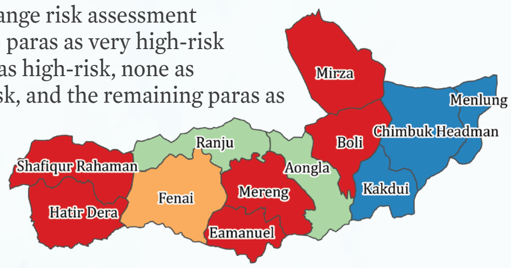
Climate Change-Induced Hazards

Eight climate hazards threaten Tangkabati 1 watershed agriculture, livelihood, ecosystem, and biodiversity. Drought, erratic rainfall, landslides, and lightning intensify across most of the paras. Flash floods, heat waves, river floods, and erosion occur at low to moderate intensity.

Climate Change Risk

Climate change risk assessment identified 6 paras as very high-risk hotspots, 1 as high-risk, none as medium-risk, and the remaining paras as low to very low-risk (figure 4).

Figure 4: Mutihazard risk map



Very High	High	Medium	Low	Very Low
6	1	0	2	3

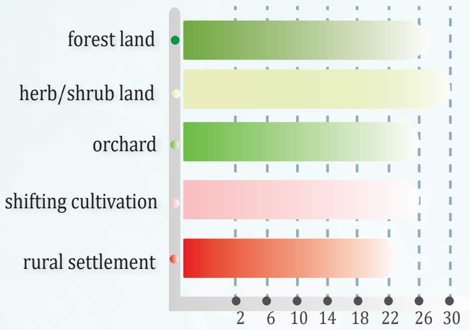
Disclaimer:
Boundaries are not necessarily authoritative

Source:
Center for Environmental and Geographic Information Services (CEGIS)

Watershed Degradation

Land Degradation Status

In B-76, land degradation is prominent in -



Causes of Degradation

- Jhum cultivation with short rotation
- Improper crop management
- Overgrazing
- Deforestation & overharvesting of trees
- Lack of vegetation cover
- Climate change
- Overuse of inorganic fertilizers

Degradation Triggering Factors

- Reduction of vegetation cover
- Quality and species composition change/diversity decline/monoculture plantation
- Soil erosion/landslides
- Surface sealing/hard pan development
- Waterlogging
- River bank erosion

Watershed Conservation

Conservation measures focused on enhancing ecological resilience as well as improving livelihood quality of the local community were identified and categorized into 6 classes.

Forest/ Shrub Land Conservation
- 20 ha Forest enrichment plantation
- 10 kilometer (km) Tree plantation along village road

Degraded Soil & Land Improvement
- 5 Landslide Treatment
- 25 Gully Treatment

Agriculture Land Conservation
- 50 ha Jhum to agroforestry
- 50 ha Fruit tree plantation
- 100 ha Orchard management
- 10 Drainage improvement

Stream Bank Protection
- <1 km Buffer strip plantation and bio-engineering methods

Water Resources Development
- 3 Irrigation canal improvement
- 0 Spring development

Capacity Development & Demonstration
- 24 Households (HHs) Regenerative farming training
- 24 HHs Drip irrigation training
- 24 HHs Homestead improvement

Watershed Management Infrastructures

Protection Wall Construction	0
Earthen Dam Construction	3
Dam Improvement	0
Irrigation Canal Construction	1
Check Dam Construction	0

Watershed conservation interventions in Tangkabati 1 include dam construction (3) and irrigation canal construction (1) being the most implemented, aimed at enhancing water storage and distribution. In addition to these, other watershed conservation measures have been identified to further boost water availability, agricultural productivity, and overall ecosystem health. Measures such as check dam construction, dam improvement, irrigation canal construction, culvert construction, and protection wall construction will help control runoff, reduce soil erosion, and protect critical infrastructure. Together, these interventions will significantly benefit the local community by ensuring year-round water access, improving crop yields, and reducing vulnerability to climate-induced hazards. These integrated efforts promote sustainable livelihoods and strengthen resilience in the Tangkabati 1 region.

ADB
JFPR Japan Fund for Prosperous and Resilient Asia and the Pacific
From the People of Japan
CEGIS